AMAZON PRODUCT SALES & CONSUMER INSIGHTS

OVERVIEW

This project analyzes e-commerce sales data to understand customer buying behavior, product performance, and revenue trends. The dataset includes orders, products, categories, sales amounts, and customer review ratings.

Through SQL analysis, it identifies monthly revenue patterns, top-performing categories, and customer satisfaction levels. These insights support better business decisions, revenue growth, and improved customer experience.

PROBLEM STATEMENT

The business lacks a structured, data-driven approach to analyze its e-commerce sales data. Although transactional data is available, it is initially stored in a flat format, making it difficult to identify revenue trends, top-performing products, customer spending behavior, and category-wise performance. Without proper normalization and analytical queries, strategic decision-making and performance optimization remain limited.

What's Happening?

- **Sales data exists but is not organized for efficient querying and analysis.**
- **There is no clear visibility into monthly revenue trends and seasonal performance.**
- **Customer lifetime value and top-spending customers are not identified.**
- **Category performance, payment method usage, and delivery patterns are not systematically evaluated.**

Solution Goal

- Normalize the raw dataset into structured relational tables (Customers, Orders, Products, Categories, Order_Items).
- Use SQL techniques such as joins, aggregations, CTEs, subqueries, and window functions to extract business insights.
- Analyze revenue trends, customer lifetime value, product rankings, and category performance.
- Generate actionable insights to improve sales strategy, customer experience, and overall business performance.

DATA DESCRIPTION

This project is based on an e-commerce sales dataset that was initially stored in a single flat table (`sales_dat`) and later normalized into multiple relational tables including Customers, Categories, Products, Orders, and Order_Items. The dataset contains transactional sales records with order details, product information, pricing, payment methods, delivery status, and customer review ratings.

The structured schema enables advanced SQL analysis such as revenue trends, customer lifetime value, product ranking, payment method performance, and category-wise sales insights.

Key Fields

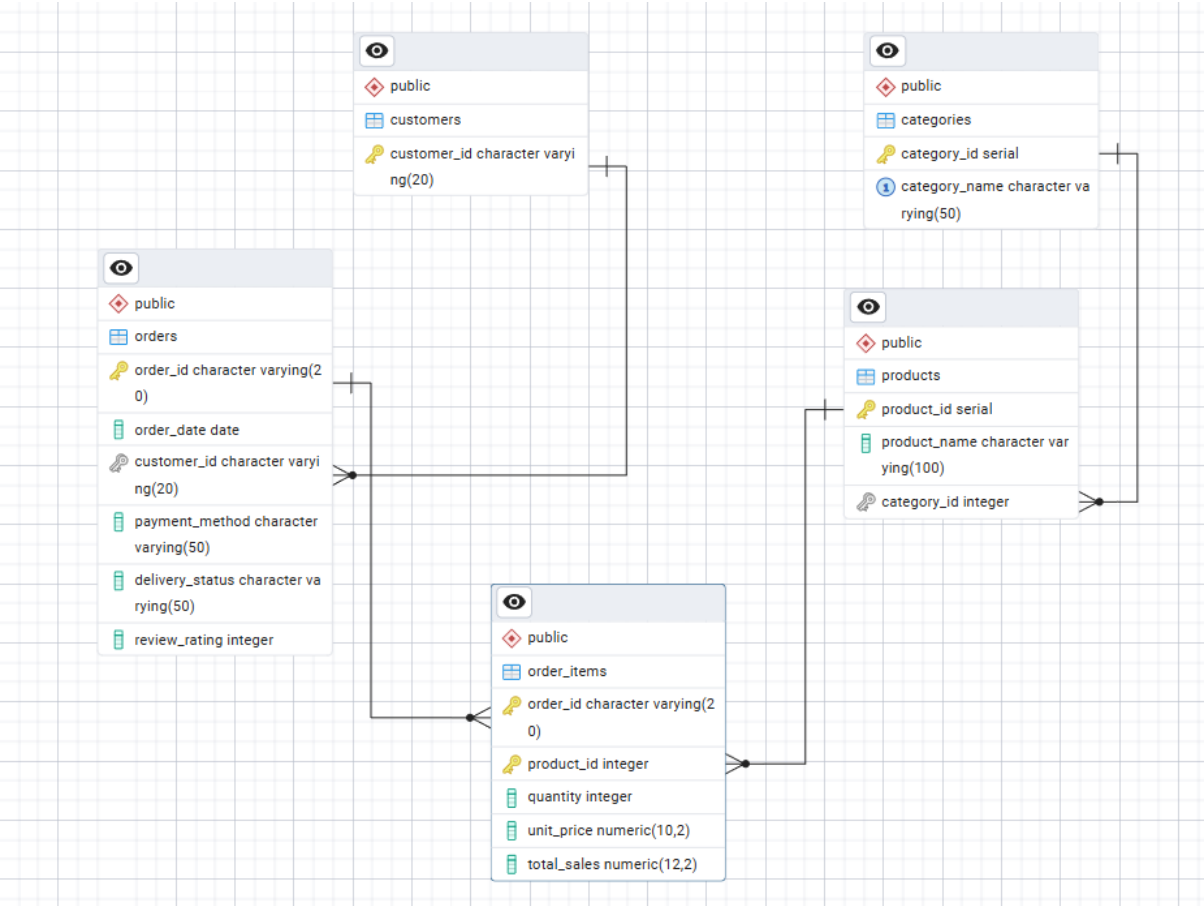
- order_id – Unique order identifier
- customer_id – Unique customer identifier
- product_id / product_name – Product details
- category_id / category_name – Product category
- order_date – Purchase date
- quantity, unit_price, total_sales – Sales metrics
- payment_method – Mode of payment
- delivery_status – Order status
- review_rating – Customer feedback score

DATA CLEANING STEPS

- Removed or handled null and missing values
- Standardized column headers and text formats
- Normalized dataset into separate SQL tables

  : (Customers, Orders, Products, Categories, Order_Items)

- Removed duplicate records
- Ensured consistent and correct data types
- Established primary and foreign key relationships for data integrity and ERD visualization



This ERD represents a normalized e-commerce database schema designed in PostgreSQL. The `customers` table stores customer information, while the `orders` table records transaction details linked to customers through `customer_id`. The `order_items` table acts as a bridge between `orders` and `products`, capturing quantity, unit price, and total sales for each product in an order. The `products` table is connected to the `categories` table using `category_id`, organizing products by category. Primary and foreign key relationships ensure data integrity and clearly define one-to-many relationships across the system.

1) Find the Total Revenue by Category

```sql
SELECT c.category_name,
SUM(oi.total_sales) as total_revenue
FROM order_items oi
JOIN products p ON oi.product_id = p.product_id
JOIN categories c ON p.category_id = c.category_id
GROUP BY c.category_name
ORDER BY total_revenue DESC;
```

	category_name character varying (50)	total_revenue numeric
1	Clothing	24405625.76
2	Electronics	22735032.58
3	Beauty	22226859.92
4	Books	19618381.22

Insight

1) The revenue analysis by category shows which product segments are driving the majority of the company's income.

2) Identifying high-performing categories helps the business allocate marketing budgets, manage inventory efficiently, and focus on profitable product lines.

3) Low-revenue categories may require promotional campaigns, pricing adjustments, or product improvements to increase overall sales performance.

2) Top 5 Customers by Total Spending (CTE)

```sql
WITH customer_spending AS (
    SELECT o.customer_id,
           SUM(oi.total_sales) AS total_spent
    FROM orders o
    JOIN order_items oi ON o.order_id = oi.order_id
    GROUP BY o.customer_id
)
SELECT *
FROM customer_spending
ORDER BY total_spent DESC
LIMIT 5;
```

	customer_id character varying (20)	total_spent numeric
1	CUST9818	403141.85
2	CUST1483	358139.52
3	CUST1981	305844.64
4	CUST5556	298485.12
5	CUST9313	298275.26

Insight

1) Identifying the top 5 customers by total spending helps the business understand who contributes the most revenue and where major profits are coming from.

2) These high-value customers are critical to business stability, so the company should focus on retention strategies such as loyalty programs, personalized offers, and premium support.

3) If a large percentage of total revenue comes from only a few customers, the business may face risk concentration, meaning losing one key customer could significantly impact overall revenue.

4) Analyzing the purchasing patterns of top customers also helps in creating targeted marketing campaigns to attract similar high-spending customers.

3)Monthly Sales Trend (Date Analysis)

```
SELECT
    TRIM(TO_CHAR(o.order_date, 'Month')) AS month_name,
    SUM(oi.total_sales) AS monthly_revenue
FROM orders o
JOIN order_items oi
    ON o.order_id = oi.order_id
GROUP BY
    TRIM(TO_CHAR(o.order_date, 'Month')),
    DATE_TRUNC('month', o.order_date)
ORDER BY DATE_TRUNC('month', o.order_date);
```

	month_name text	monthly_revenue numeric
1	January	6792065.35
2	February	7578437.49
3	March	6553777.61
4	April	9521315.27
5	May	7748187.76
6	June	9217525.07
7	July	6684806.70
8	August	7288441.03
9	September	6889814.62
10	October	6651202.24
11	November	6436737.91
12	December	7623588.43

Insight

1) The monthly revenue table helps identify seasonal sales patterns and highlights peak-performing months for the business.

2) Consistent growth in monthly revenue indicates strong demand and effective sales or marketing strategies.

3) Sudden drops in revenue during certain months may signal seasonal slowdowns, reduced customer activity, or operational challenges.

4) Comparing month-by-month performance enables management to plan inventory, promotions, and revenue targets more effectively.

4) Rank Products by Revenue (Window Function)

```
SELECT p.product_name,
       SUM(oi.total_sales) AS revenue,
       RANK() OVER (ORDER BY SUM(oi.total_sales) DESC) AS rank_position
FROM order_items oi
JOIN products p ON oi.product_id = p.product_id
GROUP BY p.product_name
limit 10;
```

	product_name character varying (100)	revenue numeric	rank_position bigint
1	Saree	6409809.61	1
2	Laptop	5948177.31	2
3	Shampoo	5331597.38	3
4	Perfume	5244685.49	4
5	Sneakers	5115418.50	5
6	Children's Book	4976939.32	6
7	T-Shirt	4846281.13	7
8	Smartwatch	4713696.14	8
9	Headphones	4346062.42	9
10	Lipstick	4343196.30	10

Insight

1) Ranking products by revenue helps the business clearly identify which products are the top performers and which ones contribute the most to overall sales.

2) High-ranking products should be prioritized for inventory planning, marketing promotions, and premium website placement to maximize revenue opportunities.

3) Low-ranking products may indicate weak demand, poor pricing strategy, or strong competition, suggesting the need for discounts, bundling strategies, or product improvements.

4) Product ranking also helps management focus on data-driven decision-making by understanding which items drive profitability and which ones may need discontinuation.

5) Highest Order Value (Subquery)

```sql
SELECT order_id, order_total
FROM (
    SELECT order_id,
           SUM(total_sales) AS order_total
    FROM order_items
    GROUP BY order_id
) sub
ORDER BY order_total DESC
LIMIT 1;
```

	order_id character varying (20)	order_total numeric
1	ORD100592	249747.75

Insight

1) Identifying the highest order value helps the business understand the maximum revenue generated from a single transaction, highlighting the presence of high-spending customers.

2) High-value orders may indicate bulk purchases, premium product sales, or successful upselling strategies, which can be replicated to increase overall revenue.

3) Analyzing these large transactions allows the company to study customer behavior, product combinations, and pricing strategies that drive bigger purchases.

4) If very few high-value orders contribute significantly to total revenue, the business may consider developing targeted campaigns to encourage more customers to place larger orders.

5) Tracking maximum order value over time also helps measure growth in customer purchasing power and overall business expansion.

6) Average Order Value

```sql
SELECT AVG(order_total) AS avg_order_value
FROM (
    SELECT order_id,
           SUM(total_sales) AS order_total
    FROM order_items
    GROUP BY order_id
) t;
```

	avg_order_value numeric
1	73179.193651315789

Insight

1) Average Order Value (AOV) measures the average amount customers spend per order, helping the business evaluate overall purchasing behavior and revenue efficiency.

2) A higher AOV indicates successful upselling and cross-selling strategies, while a lower AOV may suggest customers are purchasing fewer or lower-priced products.

3) Monitoring AOV over time helps management assess the effectiveness of pricing strategies, promotional campaigns, and product bundling offers.

4) If AOV is increasing consistently, it reflects improved customer trust and stronger buying power; if it declines, the business may need to introduce minimum order discounts or bundle deals.

5) Comparing AOV across customer segments or payment methods can reveal which groups contribute more value, enabling more targeted marketing strategies.

7) Revenue by Payment Method

```
SELECT o.payment_method,
       SUM(oi.total_sales) as total_revenue
FROM orders o
JOIN order_items oi ON o.order_id = oi.order_id
GROUP BY o.payment_method
ORDER BY total_revenue DESC;
```

	payment_method character varying (50)	total_revenue numeric
1	Cash on Delivery	24962239.69
2	Credit Card	22055228.74
3	Debit Card	21694664.54
4	UPI	20273766.51

Insight

1) Analyzing revenue by payment method helps the business understand which payment options customers prefer and which ones generate the highest sales value.

2) If one payment method contributes significantly more revenue, it indicates higher customer trust and convenience associated with that option, which the business should prioritize and optimize.

3) On the other hand, low-performing payment methods may signal usability issues, lack of customer awareness, or transaction friction, requiring technical improvements or promotional incentives.

4) Understanding payment behavior also supports strategic decisions such as partnerships with financial providers, discount offers on specific payment modes, and improving checkout experience to increase conversion rates.

8) Delivery Status Distribution

```sql
SELECT
    review_rating,

    COUNT(CASE WHEN delivery_status = 'Delivered' THEN 1 END) AS delivered,
    COUNT(CASE WHEN delivery_status = 'Pending' THEN 1 END) AS pending,
    COUNT(CASE WHEN delivery_status = 'Returned' THEN 1 END) AS returned

FROM orders
GROUP BY review_rating
ORDER BY review_rating;
```

	review_rating integer	delivered bigint	pending bigint	returned bigint
1	1	77	77	72
2	2	87	68	63
3	3	79	77	88
4	4	87	81	82
5	5	103	94	81

Insight

1) Lower ratings (1★–2★) show a higher percentage of Returned and Pending orders, indicating delivery or product issues.

2) Higher ratings (4★–5★) have a very high percentage of Delivered orders, showing strong customer satisfaction.

3) There is a clear relationship between successful delivery and positive review ratings.

4) Improving delivery efficiency and reducing returns can directly increase overall customer ratings.

9) Running Total of Revenue (Window Function)

```sql
SELECT
    o.order_date,
    SUM(oi.total_sales) AS daily_revenue,
    SUM(SUM(oi.total_sales)) OVER (
        ORDER BY o.order_date
    ) AS running_total_revenue
FROM orders o
JOIN order_items oi
    ON o.order_id = oi.order_id
GROUP BY o.order_date
ORDER BY o.order_date
```

	order_date date	daily_revenue numeric	running_total_revenue numeric
1	2025-01-01	64926.72	64926.72
2	2025-01-02	158371.04	223297.76
3	2025-01-03	348444.17	571741.93
4	2025-01-04	110864.62	682606.55
5	2025-01-05	381244.34	1063850.89
6	2025-01-06	103368.13	1167219.02
7	2025-01-07	99827.13	1267046.15
8	2025-01-08	312200.13	1579246.28
9	2025-01-09	47467.80	1626714.08
10	2025-01-10	252145.69	1878859.77

Insight

1) Running total revenue shows whether the business is growing consistently over time or experiencing slowdowns in sales momentum.

2) A steady upward trend indicates healthy and stable business expansion, while flat or slow growth may signal declining demand.

3) It helps management track progress toward revenue targets throughout the period instead of waiting for month-end results.

4) Sudden jumps in the running total can highlight successful campaigns, seasonal demand, or high-value transactions.

10) Most Frequently Purchased Product

```
SELECT p.product_name,
       SUM(oi.quantity) AS total_quantity
FROM order_items oi
JOIN products p ON oi.product_id = p.product_id
GROUP BY p.product_name
ORDER BY total_quantity DESC
LIMIT 5;
```

	product_name character varying (100)	total_quantity bigint
1	Saree	233
2	Laptop	216
3	Perfume	210
4	Shampoo	207
5	Lipstick	198

Insight

1)The most frequently purchased product reflects the highest customer demand and indicates strong market preference.

2) High purchase frequency suggests the product has consistent sales potential and should be prioritized for inventory management.

3) This product can be used in promotional bundles or cross-selling strategies to increase overall revenue.

4) If one product dominates sales volume, the business should ensure stable supply to avoid stockouts and lost sales.

11) Customer Lifetime Value with Ranking

```
SELECT o.customer_id,
       SUM(oi.total_sales) AS lifetime_value,
       DENSE_RANK() OVER (ORDER BY SUM(oi.total_sales) DESC) AS customer_rank
FROM orders o
JOIN order_items oi ON o.order_id = oi.order_id
GROUP BY o.customer_id
LIMIT 10;
```

	customer_id character varying (20)	lifetime_value numeric	customer_rank bigint
1	CUST9818	403141.85	1
2	CUST1483	358139.52	2
3	CUST1981	305844.64	3
4	CUST5556	298485.12	4
5	CUST9313	298275.26	5
6	CUST8444	292972.85	6
7	CUST1440	280396.00	7
8	CUST7773	277925.39	8
9	CUST7484	270658.31	9
10	CUST9041	269321.80	10

Insight

1) Ranking customers by lifetime value helps identify the most profitable customers who contribute the highest long-term revenue to the business.

2) High-CLV customers should be prioritized with loyalty programs, personalized offers, and premium support to ensure retention.

3) If revenue is heavily concentrated among a few top-ranked customers, the business faces dependency risk and should diversify its customer base.

4) CLV ranking supports strategic decision-making by focusing marketing efforts on acquiring and retaining high-value customers.

12) Average Review Rating by Category

```sql
SELECT
    c.category_name,
    ROUND(AVG(o.review_rating), 2) AS avg_rating
FROM orders o
JOIN order_items oi
    ON o.order_id = oi.order_id
JOIN products p
    ON oi.product_id = p.product_id
JOIN categories c
    ON p.category_id = c.category_id
GROUP BY c.category_name
ORDER BY avg_rating DESC;
```

	category_name character varying (50)	avg_rating numeric
1	Electronics	3.15
2	Beauty	3.13
3	Books	3.09
4	Clothing	3.08

Insight

1) The average review rating by category helps identify which product segments deliver the highest customer satisfaction.

2) Categories with lower ratings may indicate product quality issues, pricing concerns, or mismatched customer expectations.

3) High-rated categories build brand trust and can be promoted more aggressively to attract new customers.

4) Monitoring rating trends helps the business improve weak areas and maintain long-term customer loyalty.

RECOMMENDATIONS

- Focus marketing efforts on top-performing categories to maximize revenue growth.
- Improve performance of low-rated categories by analyzing customer feedback and addressing quality or delivery issues.
- Target high-value customers with loyalty programs and personalized offers to increase customer lifetime value.
- Monitor monthly sales trends to optimize inventory planning and seasonal promotions.
- Promote preferred payment methods to enhance customer convenience and checkout experience.

CONCLUSION

The analysis transformed raw transactional data into a structured relational database, enabling meaningful business insights through SQL queries. By evaluating revenue trends, customer spending behavior, product rankings, and review ratings, the project provides a clear understanding of sales performance and operational patterns. These insights support data-driven decision-making, helping improve revenue growth, customer satisfaction, and overall business strategy